Project: Worked examples
Title: ISO 286 limits and fits
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Hole and shaft limits for a nominal size and tolerance classes, the resulting fit (clearance, transition or interference) and its extreme values, from the ISO 286-1 formulae.

WHAT IT CALCULATES	RELATION
Standard tolerance factor at the geometric mean D of the size step	i = 0.45·D^(1/3) + 0.001·D (µm)
Width of the tolerance band for the grade	ITn = multiplier × i
Fundamental deviations by letter from the published formulae, and hole deviations by the general rule	EI = −es (A–H); ES = −ei + Δ (K–ZC)
Limits of size of the hole and the shaft	limit = nominal + deviation
Fit extremes, the largest and smallest clearance (negative is interference)	c_{max} = ES − ei; c_{min} = EI − es

Basis: Bases of tolerances, deviations and fits — tolerance grades and fundamental deviations[1](§4, §5); Tables of standard tolerance classes and limit deviations (not reproduced)[2].

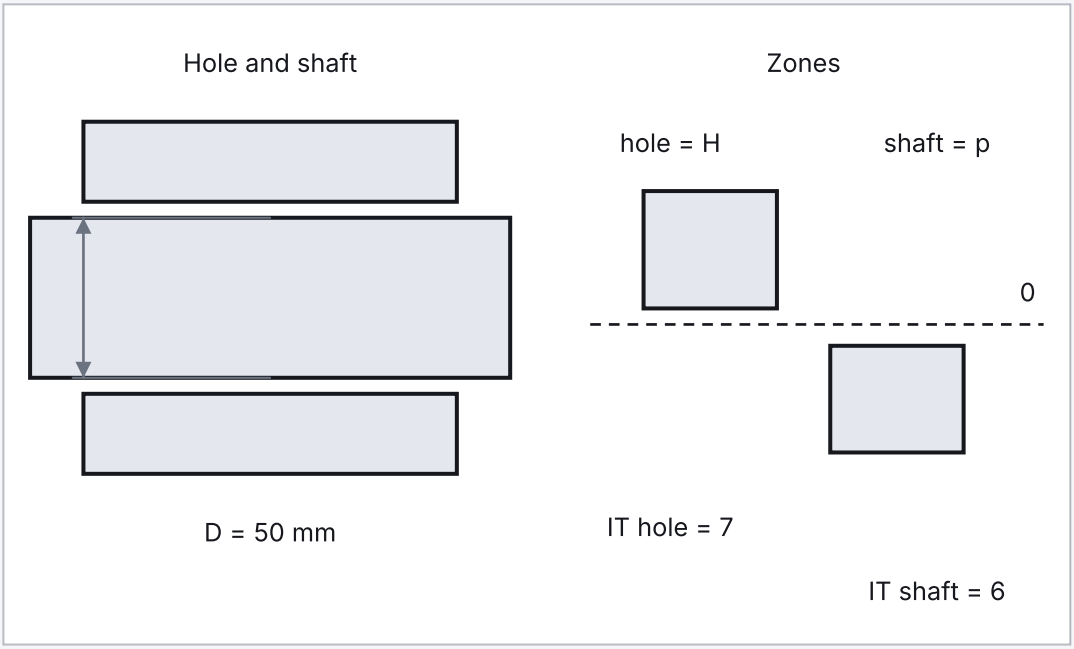


Figure 1 — A hole and its shaft of one nominal size D, and their tolerance zones about the zero line (hole above, shaft below as drawn; the letters place them). Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
D	Nominal size	50	mm	drawing: DRG-HUB-04 rev C	1 … 500 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$hole_{letter}$	Hole fundamental deviation letter	H		spec: DRG-HUB-04: H7/p6	any
$hole_{grade}$	Hole IT grade	7		spec: DRG-HUB-04: H7/p6	5 … 11
$shaft_{letter}$	Shaft fundamental deviation letter	p		spec: DRG-HUB-04: H7/p6	any
$shaft_{grade}$	Shaft IT grade	6		spec: DRG-HUB-04: H7/p6	4 … 11

3. Assumptions

A1. IT values and fundamental deviations are the unrounded results of the ISO 286-1 formulae at the geometric mean of the size step. (The tabulated values apply the standard's own rounding and may differ by about 1 µm; enter overrides from a released reference record where the tabulated value governs.)
A2. Hole deviations follow the general rule from the shaft deviations, with Δ = ITn − IT(n−1) where the standard applies it. (Table-specific departures in the standard are not reproduced; use the overrides where they govern.)

4. Working

[6] D_{step} = size step (lower, upper) [1](§4)
= (30, 50] mm
= 38.73 mm
Geometric mean of the size step

[7] i = 0.45·D^(1/3) + 0.001·D [1](§5)
= D = 38.73 mm
= 1.5612 µm
Standard tolerance unit

[8] IT_{hole} = IT7 [1](§5)
= 16 × i
= 24.98 µm
Hole tolerance

[9] IT_{shaft} = IT6 [1](§5)
= 10 × i
= 15.612 µm
Shaft tolerance

[10] es = ei + IT_{shaft} [1](§5)
= 24.98 + 15.612
= 40.592 µm
Shaft upper deviation

[11] ei = fundamental deviation 'p' [1](§5)
= +IT7
= 24.98 µm
Shaft lower deviation

[12] ES = EI + IT_{hole} [1](§5)
= 0 + 24.98
= 24.98 µm
Hole upper deviation

[13] EI = fundamental deviation 'H' [1](§5)
= −es of the same letter
= 0 µm
Hole lower deviation

[14] $D_{hole,max}$ = D + ES
= 50 mm + 24.98 µm
= 50.025 mm
Hole maximum
[15] $D_{hole,min}$ = D + EI
= 50 mm + 0 µm
= 50 mm
Hole minimum
[16] $d_{shaft,max}$ = D + es
= 50 mm + 40.592 µm
= 50.041 mm
Shaft maximum
[17] $d_{shaft,min}$ = D + ei
= 50 mm + 24.98 µm
= 50.025 mm
Shaft minimum
[18] c_{max} = ES - ei
= 24.98 µm − 24.98 µm
= 0 µm
Maximum clearance (negative: minimum interference)
[19] c_{min} = EI - es
= 0 µm − 40.592 µm
= -40.592 µm
Minimum clearance (negative: maximum interference)
[20] i_{max} = es - EI
= 40.592 µm − 0 µm
= 40.592 µm
Maximum interference
[21] i_{min} = ei - ES
= 24.98 µm − 24.98 µm
= 0 µm
Minimum interference (negative: maximum clearance)
[22] fit = fit class
= H7/p6
= "interference"
Fit kind: clearance, transition or interference
[23] c_{mean} = (c_{max} + c_{min}) / 2
= (0 µm + -40.592 µm) / 2
= -20.296 µm
Mean clearance

5. Results

SYMBOL	VALUE	DESCRIPTION
IT_{hole}	24.98 µm	Hole tolerance
IT_{shaft}	15.612 µm	Shaft tolerance
ES	24.98 µm	Hole upper deviation
EI	0 µm	Hole lower deviation
es	40.592 µm	Shaft upper deviation
ei	24.98 µm	Shaft lower deviation
$D_{hole,max}$	50.025 mm	Hole maximum

SYMBOL	VALUE	DESCRIPTION
$D_{hole,min}$	50 mm	Hole minimum
$d_{shaft,max}$	50.041 mm	Shaft maximum
$d_{shaft,min}$	50.025 mm	Shaft minimum
fit	"interference"	Fit kind
c_{max}	0 µm	Maximum clearance
c_{min}	-40.592 µm	Minimum clearance
i_{max}	40.592 µm	Maximum interference
i_{min}	0 µm	Minimum interference
c_{mean}	-20.296 µm	Mean clearance

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Fit classified	H7/p6	interference	0	

7. Applicability

No warnings.

8. Verification

Template mech.iso286-fit@1.0.1 Content hash f8c5789f0d6f6364 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. International Organization for Standardization. Geometrical Product Specifications (GPS)—ISO Code System for Tolerances on Linear Sizes—Part 1: Basis of Tolerances, Deviations and Fits. ISO 286-1:2010. ISO; 2010.
2. International Organization for Standardization. Geometrical Product Specifications (GPS)—ISO Code System for Tolerances on Linear Sizes—Part 2: Tables of Standard Tolerance Classes and Limit Deviations for Holes and Shafts. ISO 286-2:2010. ISO; 2010.